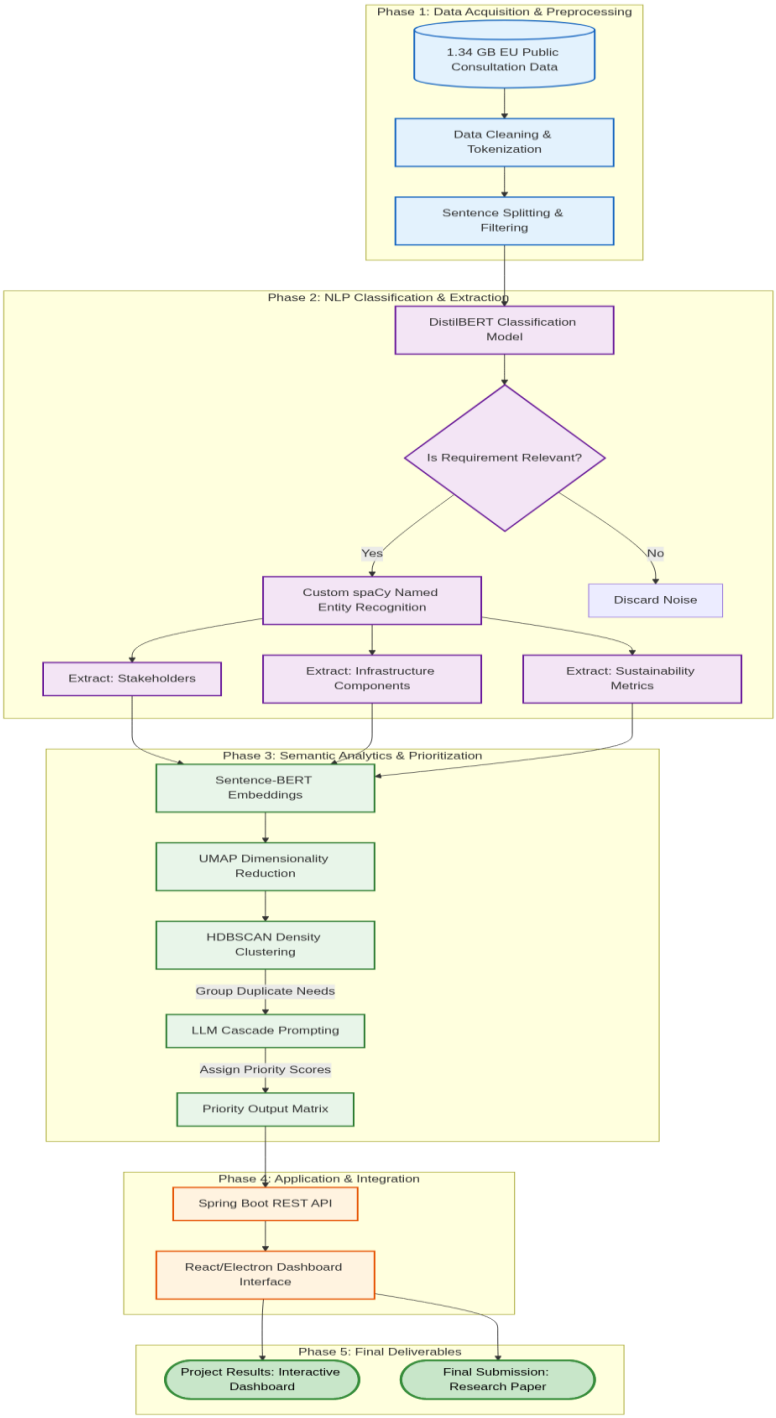
Shivang Mangal **Devanshi Jain**

SPHERE Project Timeline and Workflow with the work distribution

1. System Architecture & Workflow



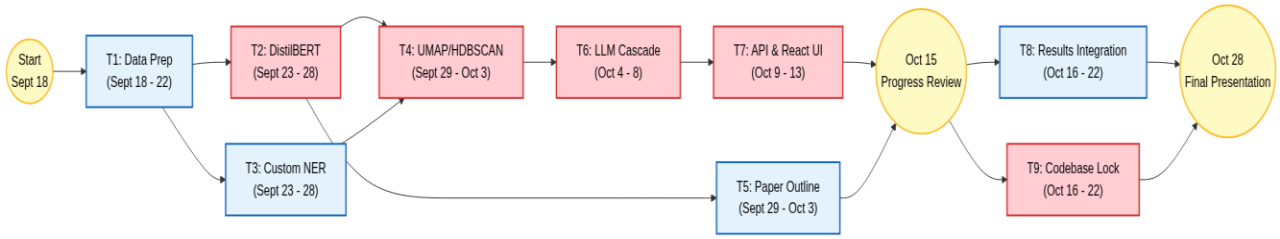
Shivang Mangal **Devanshi Jain**

2. Fast-Track Timeline (PERT Chart)

Absolute Deadline:
Progress Review: 15th Oct 2026

Final Presentation: 28th Oct 2026

The following PERT chart outlines task dependencies, estimated durations, and critical paths (highlighted in red).



3. Work Division

Shivang (Data & Infrastructure Lead)

- Dataset Management: Ingesting and preprocessing the 1.34 GB Zenodo dataset.
- Core Classification & Backend: Fine-tuning the DistilBERT noise filter and developing the Spring Boot REST API.
- Paper Contribution: Drafting sections on Dataset Acquisition, Preprocessing, Backend Architecture, and Transformer Classification methodology.

Devanshi (Analytics & Frontend Lead)

- Advanced Extraction: Adapting the custom spaCy NER model for civil engineering entities and tuning the UMAP/HDBSCAN clustering algorithms.
- Prioritization & UI: Implementing the LLM priority cascade and building the React/Electron dashboard interface.
- Paper Contribution: Drafting sections on Entity Extraction, Mathematical Clustering, Prioritization matrices, and the Literature Survey.